

CONTINUOUS ASSESSMENT 7

Multivariate Analysis

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Abstract

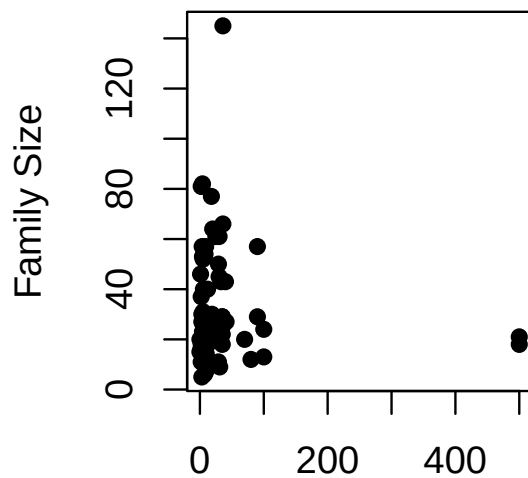
A brief glance at PCA

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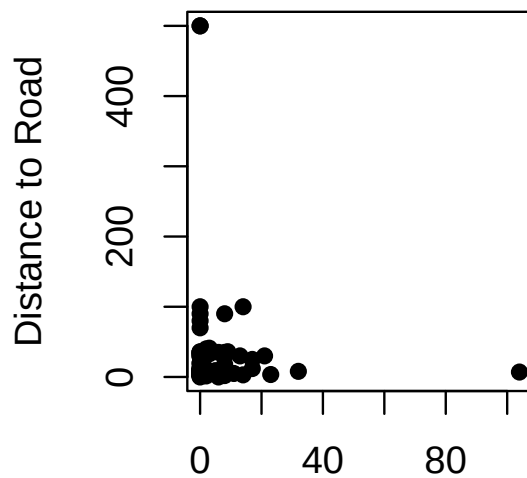
Question 1

**Family Size vs
Distance to Road**



Distance to Road

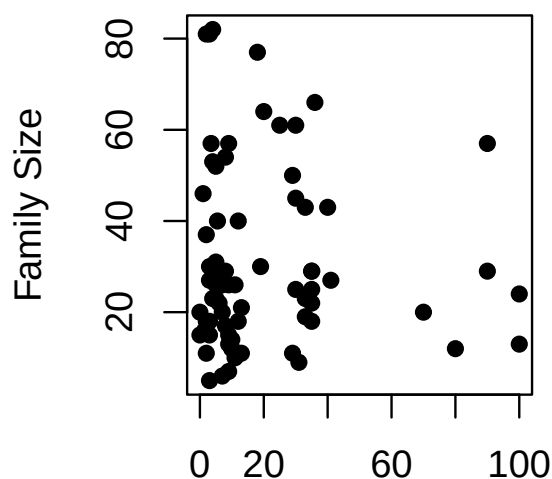
**Distance to Road vs
Cattle**



Number of Cattle

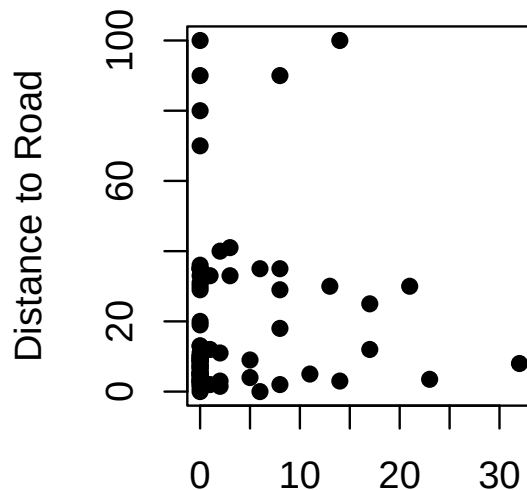
After removing the outlier the graphs look like this.

**Family Size vs
Distance to Road**



Distance to Road

**Distance to Road vs
Cattle**



Number of Cattle

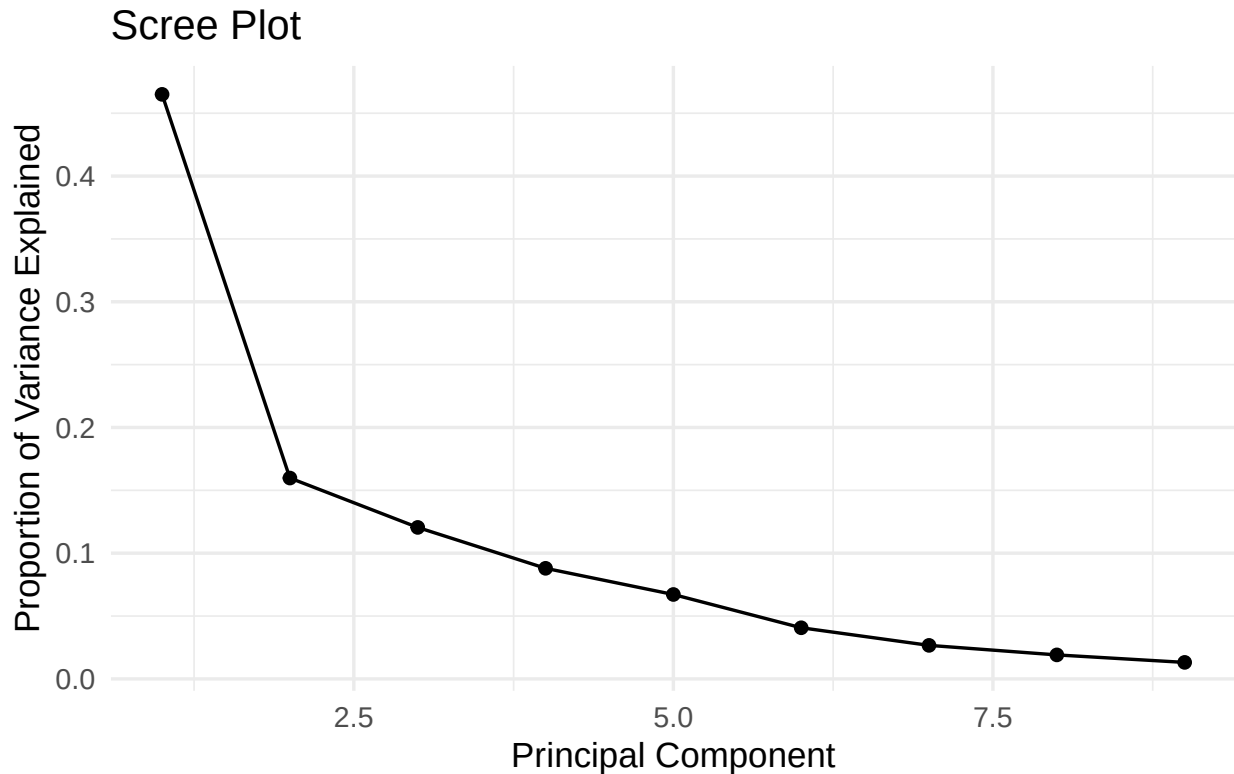
Question 2

We need to standardise the data prior to performing PCA as the variables in this data set are all measured in very different units and on different scales. Once we standardise the data it ensures that no single variable dominates the PCA owing to its scale rather than its underlying variation.

Question 3

We now perform PCA using the eigen decomposition of the covariance matrix. We first calculate the covariance matrix of the data, then we perform the eigen decomposition to obtain the eigenvalues and eigenvectors. The eigenvalues represent the amount of variance explained by each principal component, and the eigenvectors represent the direction of the principal components in the original variable space. We report the first two principal components, which capture the most variance in the data.

We now look at the scree plot to determine how many principal components to retain. The scree plot shows the proportion of variance explained by each principal component, we see that the first two principal components explain a significant amount of variance in the data. The remaining components each contribute negligibly and are therefore discarded. We retain two principal components to summarise the data



Question 4

We now perform PCA using an SVD approach. We first need to standardise the data, then we can perform the SVD and extract the principal components.

We will only report the scores for the first two principal components, which capture the most variance in the data though we have verified that the all the PCs match between the two methods.

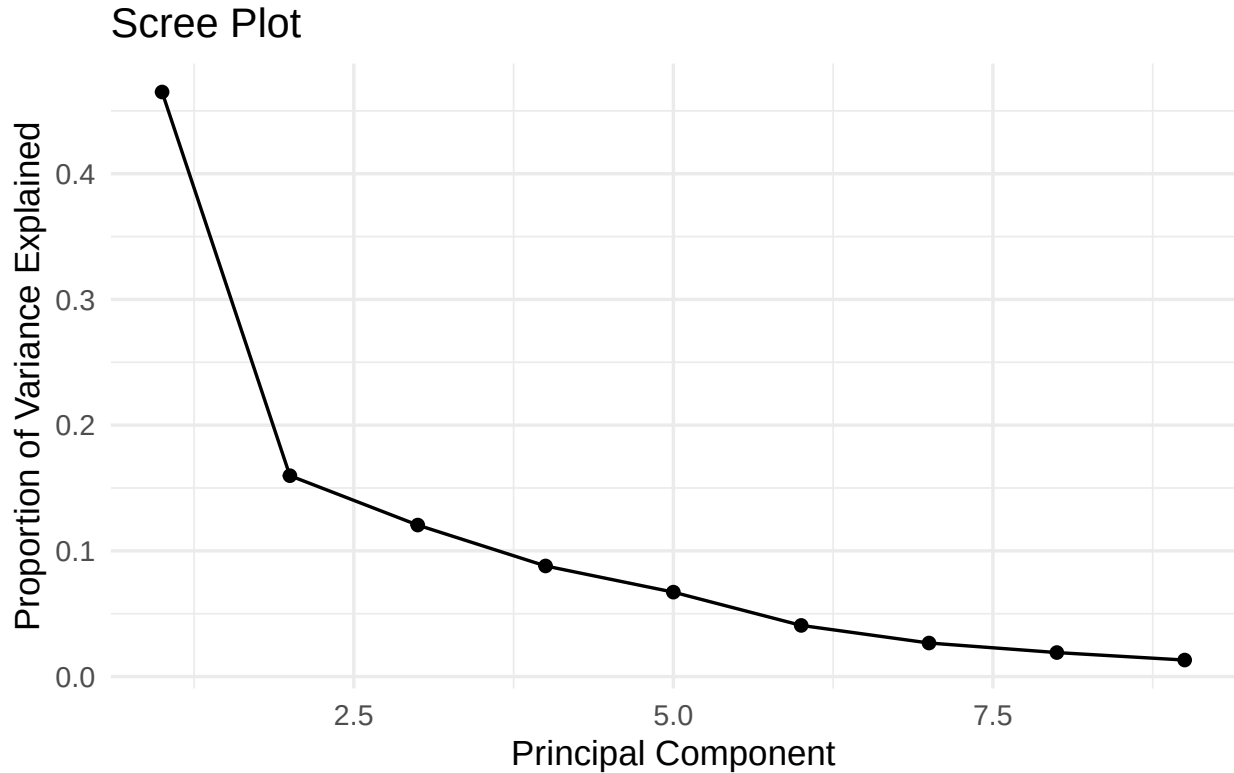
Table 1: Comparison of Principal Component Scores from Eigen Decomposition and SVD Methods

PC1_Eigen	PC2_Eigen	PC1_SVD	PC2_SVD
-1.304	-0.998	-1.304	-0.998
3.389	-2.289	3.389	-2.289
-2.277	-0.601	-2.277	-0.601
-0.910	-0.506	-0.910	-0.506
1.869	-2.615	1.869	-2.615
-1.073	-0.979	-1.073	-0.979
-1.518	-1.194	-1.518	-1.194
-1.022	-0.741	-1.022	-0.741
1.342	-1.468	1.342	-1.468

PC1_Eigen	PC2_Eigen	PC1_SVD	PC2_SVD
-0.597	-0.858	-0.597	-0.858
-1.819	-0.058	-1.819	-0.058
-1.533	-0.056	-1.533	-0.056
-1.570	0.700	-1.570	0.700
-0.650	0.747	-0.650	0.747
1.936	1.397	1.936	1.397
-1.391	0.140	-1.391	0.140
-0.769	0.404	-0.769	0.404
-2.166	0.236	-2.166	0.236
2.305	-1.010	2.305	-1.010
-1.414	0.026	-1.414	0.026
-1.760	0.176	-1.760	0.176
-0.350	0.483	-0.350	0.483
1.605	-0.605	1.605	-0.605
-0.150	1.769	-0.150	1.769
-1.849	0.379	-1.849	0.379
-1.001	0.508	-1.001	0.508
-0.786	0.923	-0.786	0.923
0.610	1.402	0.610	1.402
5.305	2.415	5.305	2.415
-1.953	0.733	-1.953	0.733
-0.400	1.419	-0.400	1.419
-0.793	0.564	-0.793	0.564
-1.818	0.626	-1.818	0.626
0.187	-0.003	0.187	-0.003
-1.265	0.488	-1.265	0.488
2.083	0.756	2.083	0.756
-2.337	-0.205	-2.337	-0.205
-1.378	0.501	-1.378	0.501
-1.136	-0.390	-1.136	-0.390
1.284	1.066	1.284	1.066
-1.230	0.058	-1.230	0.058
1.600	-0.368	1.600	-0.368
4.201	-0.272	4.201	-0.272
-1.525	0.801	-1.525	0.801
-1.920	0.519	-1.920	0.519
0.140	0.566	0.140	0.566
-1.133	0.628	-1.133	0.628
-0.794	1.135	-0.794	1.135
-1.149	0.740	-1.149	0.740
6.842	3.726	6.842	3.726
-1.486	0.364	-1.486	0.364
-0.241	0.468	-0.241	0.468
-2.411	0.122	-2.411	0.122
3.229	0.407	3.229	0.407
4.413	-0.106	4.413	-0.106
0.866	-0.011	0.866	-0.011
-0.497	1.057	-0.497	1.057
-0.092	0.137	-0.092	0.137
0.320	-0.265	0.320	-0.265
4.478	-2.938	4.478	-2.938
-0.743	0.282	-0.743	0.282
-1.487	-0.831	-1.487	-0.831
0.164	-0.632	0.164	-0.632
0.509	-0.538	0.509	-0.538

PC1_Eigen	PC2_Eigen	PC1_SVD	PC2_SVD
-1.088	0.376	-1.088	0.376
-1.408	-0.331	-1.408	-0.331
-0.197	0.635	-0.197	0.635
3.940	1.163	3.940	1.163
-1.932	-2.049	-1.932	-2.049
0.676	-3.226	0.676	-3.226
-0.712	-1.488	-0.712	-1.488
3.743	-3.411	3.743	-3.411

We also check that the scree plot matches the one obtained from the eigen decomposition method.



Question 5

This is what our loadings look like for PC1 and PC2

Table 2: A table showing the different loadings for the first 2 principle components

Variable	loadingPC1	loadingPC2
family	0.434	0.065
dist_rd	0.008	-0.497
cotton	0.446	-0.009
maize	0.352	-0.353
sorg	0.204	0.604
millet	0.240	0.415
bull	0.445	-0.068
cattle	0.355	-0.284
goats	0.255	0.049

PC1 can be interpreted as a general measure of household wealth, as it loads heavily on cotton (0.446), bull (0.445), and family (0.434) and is positive across all variables, it barely has a loading for distance to the

road which means that house hold wealth is not dependent on a households distance to a road (geographical location).

PC2 contrasts subsistence-oriented farming (sorghum and millet loading positively) against market integration and geographic accessibility (distance to road, maize and cattle loading negatively). The `dist_rd` could indicate market integration as being closer to a road allows for easier access to market stalls and transportation of crops. Thus a small distance would increase a households food security.

Question 6

We now compute the correlation between the original variables and the first principal component scores. This will help us understand which original variables contribute most to the first principal component.

Table 3: Correlation between Original Variables and the First Principal Component

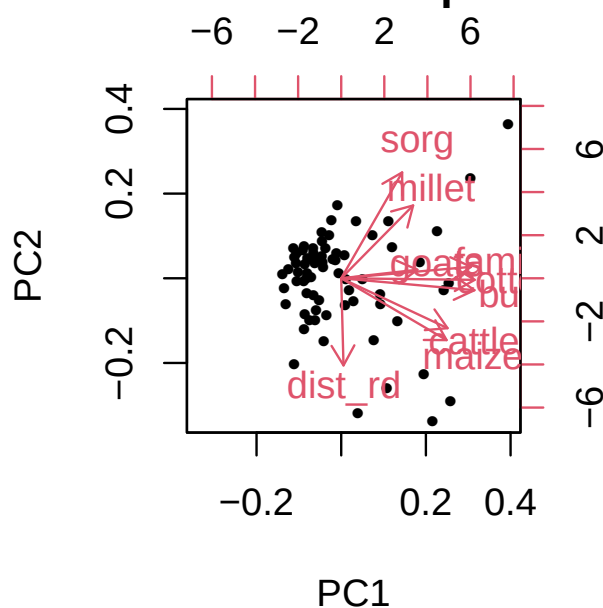
	Variable	Correlation_with_PC1
family	family	0.8875378
dist_rd	dist_rd	0.0155212
cotton	cotton	0.9126957
maize	maize	0.7205745
sorg	sorg	0.4165618
millet	millet	0.4917210
bull	bull	0.9109228
cattle	cattle	0.7270865
goats	goats	0.5207471

The correlations between PC1 and the original variables closely mirror the PC1 loadings. Variables with large positive loadings (cotton, bull, family) show the highest correlations with PC1, while variables with low loadings (dist_rd) have the lowest correlations.

Question 7

We now produce a biplot of the first two principal components.

Biplot of First Two Principal Components



The biplot above shows that most (excluding distrd) of the variables point in the same positive direction of PC1 and thus are correlated. Along PC2 sorg and millet point upwards while cattle, maize and distrd all point downwards. the other variables are quite uncorrelated and thus orthogonal to PC2. This creates two groups one for positive (subsistence) and one negative (market integration). The observations are heavily clustered near the origin, suggesting that most households in the Sikasso region are relatively homogeneous. There are a few outliers especially for PC2.